A person is seen from behind, sitting at a desk in a dark room. They are looking at several computer monitors. The monitors display various data, including what appears to be a world map and some charts or graphs. The room is dimly lit, with the primary light source being the screens. The overall atmosphere is one of focused work in a technical or security environment.

Security Operations and Management

Learning Objectives

Understand how SOC contributes in an organization's security management in its efforts to maintain good security posture of the organization. This module will help you understand the pivotal role of People, Technology, and Processes in SOC operations. It will also emphasize upon the role of SOC analysts (L1 and L2) and their coordination with other essential people in the SOC for effective threat detection and response. It will help you build knowledge around fundamentals of SOC including SOC capabilities, SOC Model Types, SOC Operations, SOC Components, SOC Workflow, etc..

The learning objectives of this module are as follows:

01

Understand the Principles of Security Management and Identify the Role of Security Operations in Effective Security Management

02

Discuss Security Operations Center (SOC) and Analyze Its Importance, Capabilities, and Functions

03

Describe the SOC Workflow and Identify the People, Processes, and Technologies Involved in SOC

04

Compare Different SOC Models and their Respective Advantages and Disadvantages

05

Explain the Concept of SOC Maturity Models and Evolution of SOC

06

Identify KPI's and Challenges and Implement Best Practices for Effective SOC Operations and Management

Objective

01

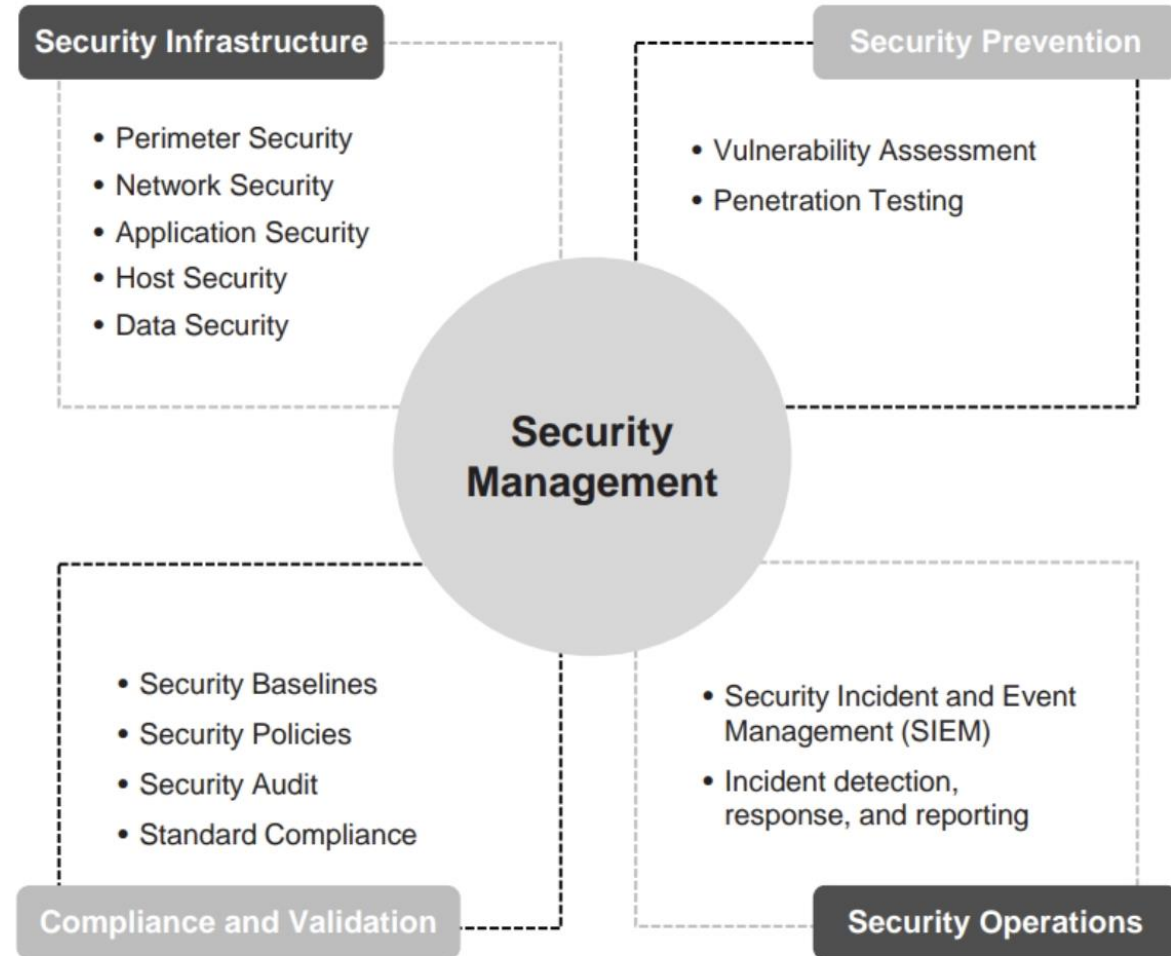
**Understand the Principles of Security
Management and Identify the Role of
Security Operations in Effective Security
Management**

Security Management

- Security management is a set of **security activities executed** by organizations to maintain their security posture at an adequate level

Different Security Activities Involved in Security Management:

- **Security Infrastructure:** Implementing adequate information **security controls**
- **Security Prevention:** Assuring security through **vulnerability management** and **penetration testing**
- **Compliance and Validation:** Complying with and validating various standards
- **Security Operations:** **Monitoring, detecting,** and **responding** to incidents





Preventing cyber attacks reliably is almost impossible due to ever-growing number of sophisticated and advanced attack techniques. So, organizations need to turn their attention from preventing attacks to rapid threat detection and response

Security Operations

- Security operations is **continuous operational practice** for maintaining and managing a secure IT environment through the implementation and execution of certain services and processes
- A predefined set of processes and services are to be followed during the daily security operation tasks based on the organization's security baselines
- In recent security operations, organizations are incorporating a third aspect of security operation known as situational awareness along with two traditional aspects of the security operations: security monitoring and security incident management
 - **Situational Awareness:** Threat intelligence can play a vital role in creating situational awareness, making informed security decisions, and shaping cyber defenses accordingly
 - **Security Monitoring:** Collecting, storing, and analyzing logs/data from different security devices to identify security incidents
 - **Security Incident Management:** Resolving security incidents with minimal adverse impact
- A dedicated unit is established by the organizations to handle and manage their security operations known as **Security Operations Center (SOC)**

Objective 02

Discuss Security Operations Center (SOC) and Analyze its Importance, Capabilities, and Functions

Security Operations Center (SOC)

01

SOC is a **centralized unit** that continuously monitors and analyzes the ongoing activities on an organization's information systems, such as networks, servers, endpoints, databases, applications, websites, etc

02

SOC **provides a single point of view** through which an organization's assets are monitored, assessed, and defended from threats

03

SOC evaluates an organization's security posture for any anomalies in its assets or information systems

04

SOC facilitates **situational awareness** and **real-time alerting**, if any type of intrusion or attack is detected

Proactive inspection of organization resources

Shielding and eradication of threats

Alert notification and compliance management

Defense mechanism against threats

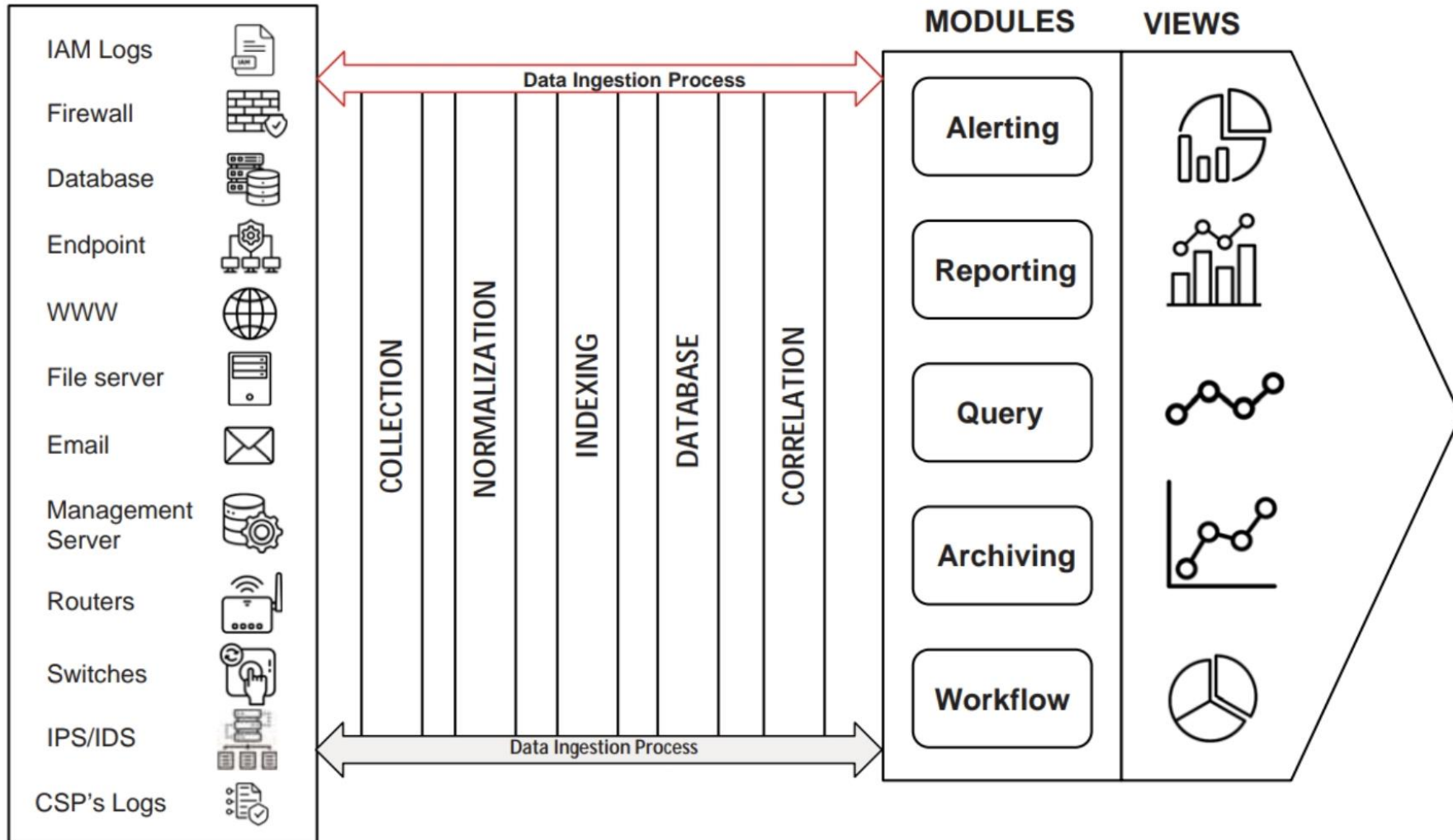
Record creation and root cause investigation

Creation of security strategy

Functions of Security Operations Center (SOC)

Security Operations Center (SOC) (Cont'd)

DATA TYPES \ DATA SOURCES



Security
Operations
Center (SOC)

Need for Security Operations Center

- To research and identify **potential cyber threats**
- To manage and handle numerous devices that are **no longer managed, unsupported, or legacy**
- To track and **secure an organization's data** and **information** around the clock
- To monitor and manage **isolated** and **localized** point solutions devices
- To monitor, mitigate, and manage **privileged user abuse**
- To respond faster to **cyber threats**
- To minimize **business loss**
- To monitor continuously for effective **incident detection**
- To implement **tools** that fit an organization's requirements
- To have **full control** over staff resources
- To customize **information security tools**
- To have full control over an organization's log data for **analysis** and **compliance**
- To reduce **security vulnerabilities**

SOC Capabilities

Situational Awareness Deliverance

- SOC can provide information regarding the on-going activities across different parts of the IT infrastructure
- It does this by **aggregating** and **associating** all the data streams that are produced by numerous devices

Threat Control and Prevention

- SOC not only improves its **imposed defenses**, but also utilizes its external resources to remain updated
- This enables SOC to provide **threat control** and **prevention**

Forensics

- SOC analysts use **structured log data** to conduct investigation for identifying the root cause of a particular attack pattern and restrict attacker's ability to perform attacks against organization

Audit and Compliance Support

- SOC not only collects and stores logs, but also **retrieves** them in an efficient manner when preparing for an audit

SOC Operations

Log Collection

- Logs are collected from **various devices on a network**, which can impact **an organization's security**

Log Retention and Archival

- Collected logs are **recovered** and **stored centrally**
- They can be used to perform **forensics** as well as **threat control** and **prevention**

Log Analysis

- Logs are analyzed through SOC's technology SOC's analyze the logs to extract important information, such as relevant **metrics** from the raw data

SOC Operations (Cont'd)

Monitoring of Security Environments for Security Events

- Information received by log analysis is transferred to the SOC team for **monitoring purposes**, so that it can be used to identify the current security position of an organization

Event Correlation

- The events from the various sources are **correlated and contextualized** based on a set of predefined correlation rules

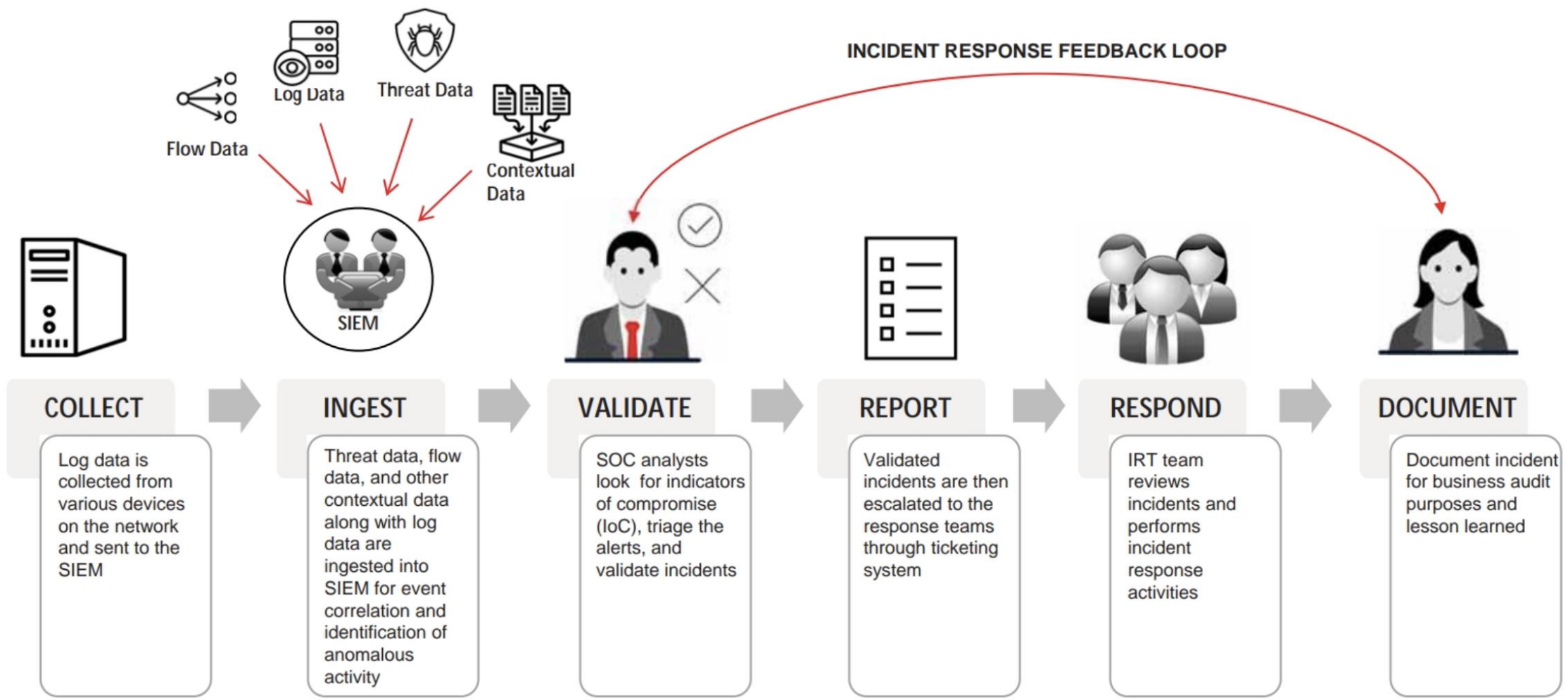
Incident Management

- It is the process of utilizing SOC's resources in an efficient manner
- This is performed by **prioritizing the incidents** based on predefined rules and objectives

Objective 03

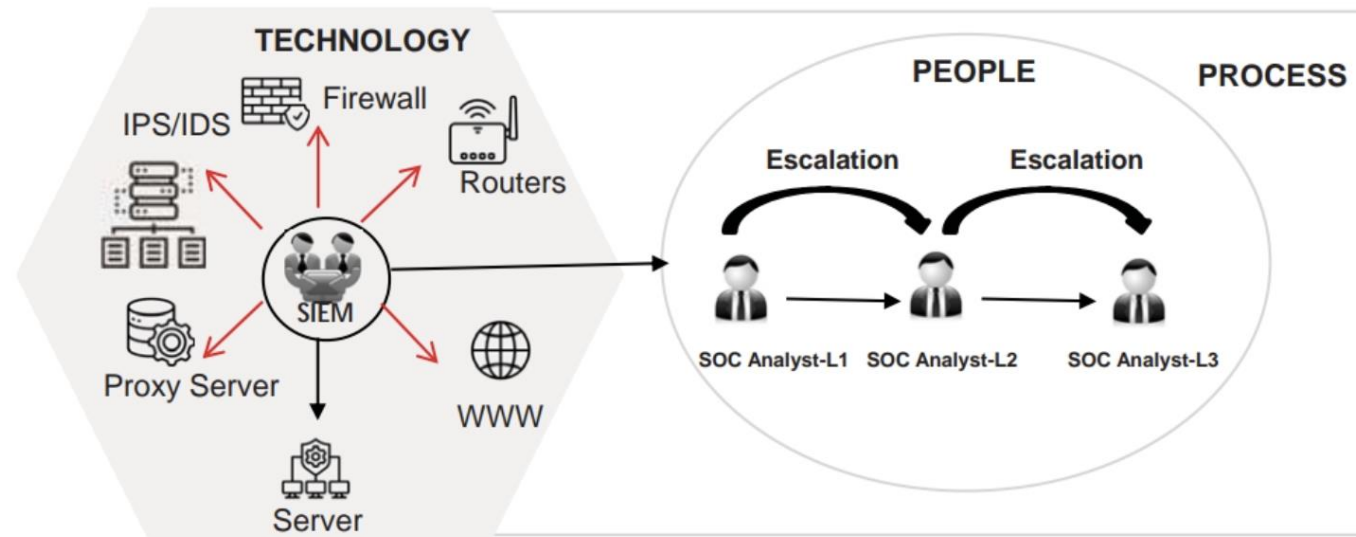
**Describe the SOC Workflow and Identify
the People, Processes, and Technologies
Involved in SOC**

SOC Workflow



Components of SOC: People, Processes, and **Technology**

- A SOC requires cooperation and communication among **People** (Analysts, Operators, Administrators, Engineers, etc.) who monitor and analyze an organization's IT infrastructure using a combination of **Processes, Procedures, and Technology**
- Skilled people for defined processes should have proper knowledge of intelligence technologies
- Processes that are planned according to the technology should act as a **connection between people and the technology**. They should ensure that both people and technology are operating adequately



Components of Security Operations Center (SOC)

Security Operations Center: **People**

- 1 People are the **specialized individuals** working at different levels of SOC
- 2 They should have **deep technical knowledge**, **wide range of capabilities**, and **variety of experiences**
- 3 They should be able to **monitor** and **analyze** large amounts of data/information that can be used for further investigations
- 4 They should possess necessary **training** and **certifications** required to fulfill their respective roles and responsibilities

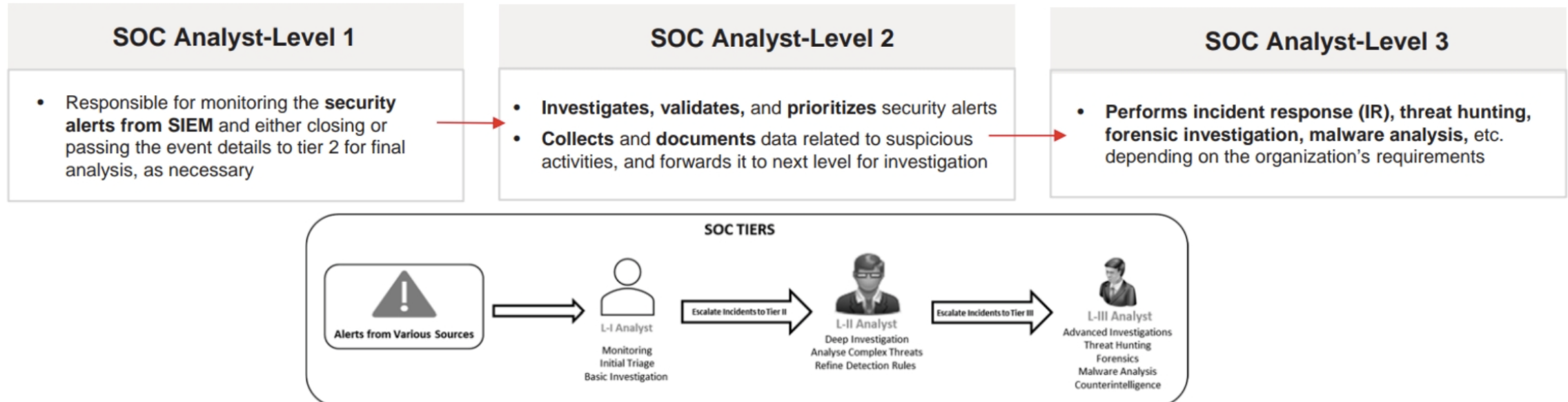
People: SOC Analyst

- The Security Operations Analyst, also known as **SOC Analyst** or **Security Analyst**, plays a pivotal role in SOC

SOC Analyst

- **Monitor** and **triage** thousands of alerts and events from various sources
- Perform **initial analysis** on **alerts** and **events** to validate and prioritize the most important
- Determine the **alerts** that require escalation to the incident response (IR) for further in-depth analysis and remediation

SOC Analyst Job Roles and Responsibilities are Categorized into Three Different Levels



People: Other Job Roles in SOC

Threat Hunters / Subject Matter Experts

- Responsible for proactively searching for signs of advanced threats within the network
- Research outside the organization by analyzing publicly available threat intelligence
- Responsible for developing, tuning, and implementing threat detection analytics
- Should have in-depth knowledge on advanced network forensics, intrusion detection, and cyber incident response

Vulnerability Analysts

- Responsible for hunting down security weaknesses and then assessing, reporting, fixing, and managing those risks across all systems

Malware Analysts

- Responsible for reverse engineering malware to understand its behavior and capabilities
- Provide threat intelligence to the SOC and improve future detection and response efforts

SOC Manager

- Responsible for managing different **resources**; **coordinating** and **communicating** with the management; and **directing** the SOC team as per the planned security strategy

Forensic Analysts

- Responsible for investigating security incidents through digital forensic analysis
- Examine and analyze a threat's structure, components, source, purpose, and the extent to which it has infiltrated and affected business systems

Compliance Analysts

- Responsible for ensuring adherence to regulatory and industry security standards
- Helps to standardize processes and Oversee compliance protocols

Threat Intelligence Analysts

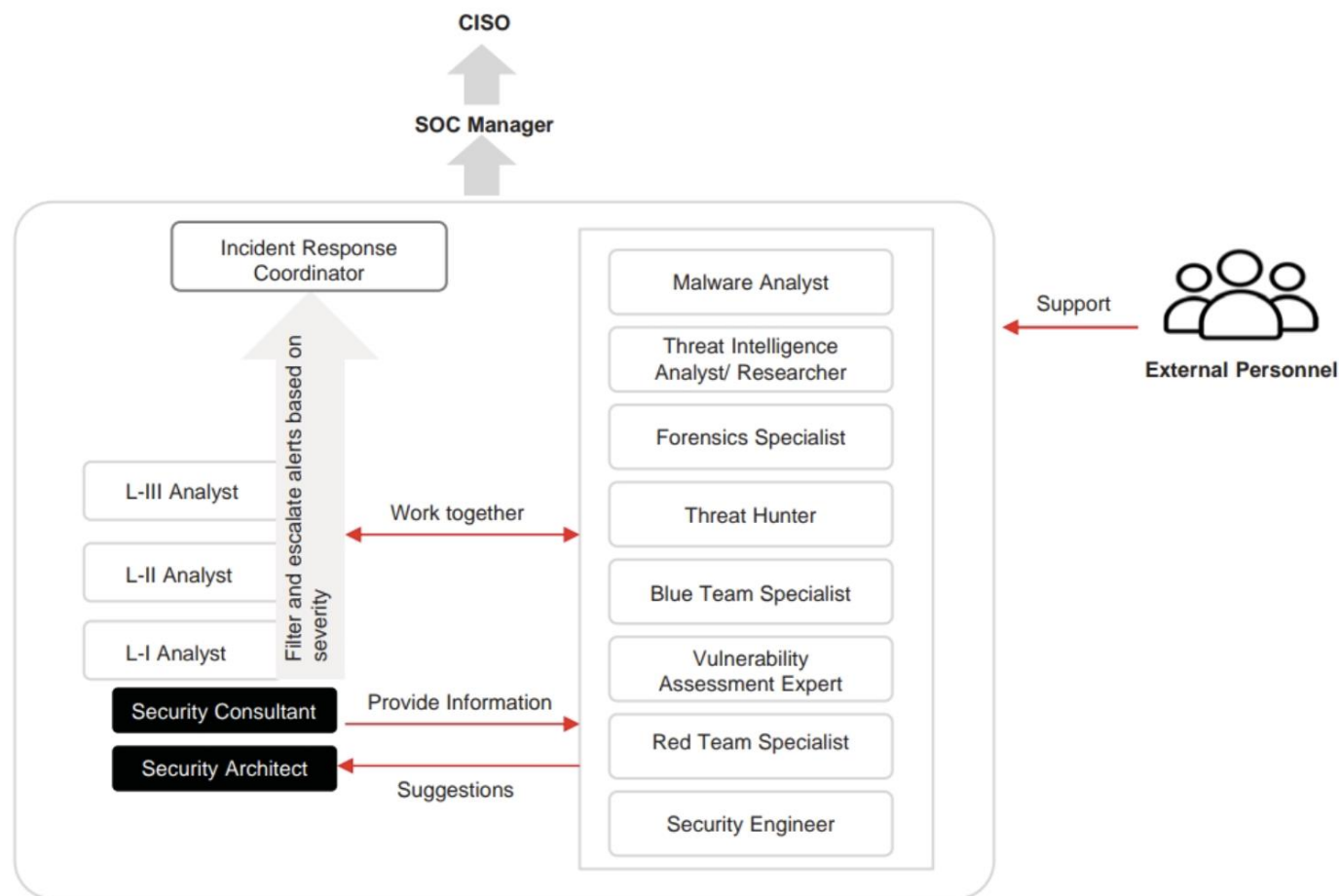
- Responsible for researching threats, analyzing threat actor behavior, and providing insights for proactive defense

Chief Information Security Officer (CISO)

- Responsible for **finalizing strategies, policies, and procedures** regarding all aspects of cybersecurity

Collaboration and Communication of Different Roles within SOC

- Security operations center (SOC) ensures robust cybersecurity through dynamic collaboration and effective communication among specialized roles
- Seamless information sharing and clear escalation paths throughout the process enable efficient detection, analysis, and response to threats, thereby protecting organizational assets and minimizing potential damage



Collaboration and Communication of Different Roles within SOC

Security Operations Center: **Processes**

1

Processes are used by the different functional parts of the SOC to perform **seamless** and **effective operations**

2

They behave as a **link** between people and technology

3

The right team will perform the **right tasks** through **well-defined process**

Security Operations Center: **Processes** (Cont'd)

Typical Processes involved in SOC

Business Processes

- In these processes, administrative components are defined and documented for efficient functioning of SOC
- They position the operations as per the organizational objectives
- **Examples:** Report preparation, log retention, etc

Operational Processes

- These processes describe the different activities that are performed in a SOC
- **Examples:** Shift scheduling, employee training

Technology Processes

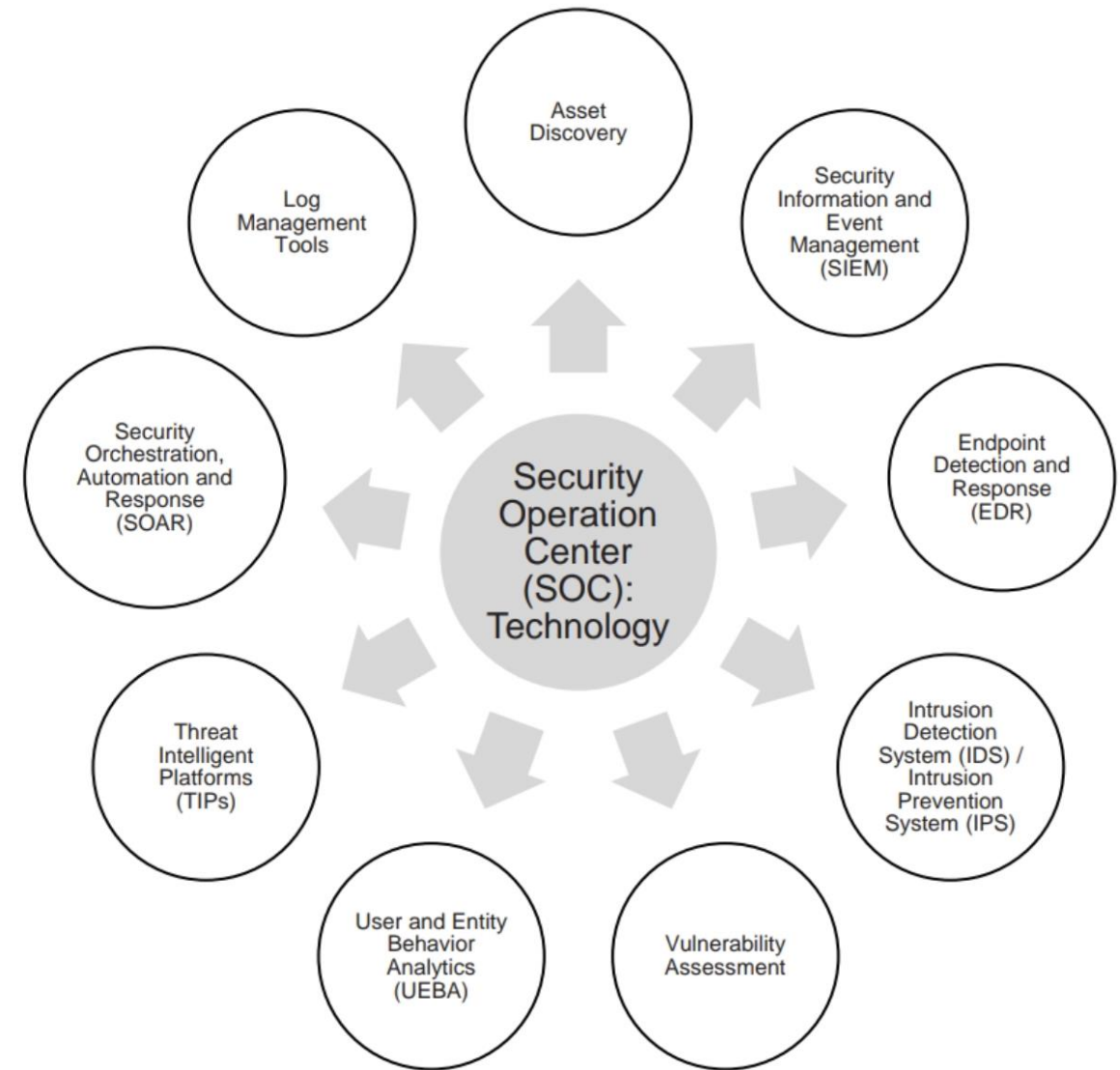
- In these processes, actions related to IT infrastructure is defined and documented
- They ensure that IT infrastructure will work at best levels at any particular time
- **Examples:** Vulnerability scanning and remediation, firmware, etc

Analytical Processes

- Analytical processes explain the way to detect and remediate security issues
- They include different methods of identifying and understanding surfacing threats
- **Examples:** Incident classification, detection and escalation, ticketing, and forensics

Security Operations Center: Technology

- Technology utilized within a Security Operations Center comprises a range of tools and systems that monitor, detect, analyze, and respond to security threats and incidents, all working together to ensure effective protection of systems and networks while facilitating seamless collaboration



Different Technologies of Security Operations Center (SOC)

Training and Managing SOC Staff

Training needs for personnel within a security operations center (SOC) are critical to ensure that the team is effective in identifying, assessing, and responding to cybersecurity threats

Key Factors for Training SOC Staff

1

Cultivating Expertise Through Training and Mentorship

- **Continuous Learning:** Regular training and mentorship ensure that the teams stay updated on cyber threats

2

Establishing Clear Security Policies

- **Standardized Procedures:** Clear security policies outline organization-wide protocols
- **Collaborative Approach:** Policies developed with inputs from various departments ensure comprehensiveness

3

Setting Measurable Goals and Priorities

- **Collaborative Goal Setting:** SOC involvement in identifying critical focus areas enhances goal-setting
- **Focused Priorities:** Targets may include improving incident response times or enhancing threat detection

4

Overseeing SOC Activities with a Data-Driven Approach

- **Performance Monitoring:** Regularly review metrics to gauge team effectiveness
- **Data-Driven Improvement:** Utilize data to pinpoint areas for training and process optimization

5

Optimizing SOC Tools and Resources

- **Staying Up-to-Date:** Continuous evaluation of security tools enhances capabilities
- **Resource Optimization:** Ensure adequate budget, training, and staffing for tool utilization

Training and Managing SOC Staff **Do's and Don'ts**

It is essential to follow best practices to ensure efficiency, collaboration, and preparedness. Below is a list of do's and don'ts for effective SOC staff management

Do's:

- Invest in comprehensive training
- Encourage continuous learning
- Promote collaboration and knowledge sharing
- Regularly conduct drills and exercises
- Provide constructive feedback
- Establish clear career paths
- Emphasize soft skills
- Implement rotation programs
- Stay ahead of emerging threats
- Encourage feedback from staff

Don'ts:

- Neglect training and development
- Micromanage
- Dismiss soft skills
- Rely solely on formal training
- Ignore burnout
- Overlook cross-training opportunities
- Neglect the importance of feedback

Objective 04

**Compare Different SOC Models and their
Respective Advantages and
Disadvantages**

Types of SOC Models

- The selection of a specific type of SOC model depends upon the **requirements, processes, and day to day functionalities of an organization**

Three Different Types of SOC Models

1. In-House/ Internal SOC Model

- An in-house/internal SOC model is recommended for those organizations that have **security issues** related **to outsourcing**
- Outsourcing affects the **integrity** and **functionality** of a business

Advantages

- It helps the in-house staff understand the organization and its environment in a much better manner, as compared to third-party security service provider
- It provides the complete picture with respect to the security posture of an organization

Disadvantages

- This model takes **many years** to set up infrastructure, threat intelligence, and other capabilities
- It requires **huge advance investment**

Types of SOC Models (Cont'd)

2. Outsourced SOC Model

- It provides a **robust security** solution to an organization
- In this model, Managed Security Service Provider (MSSP) sets up the infrastructure and offers **threat intelligence** and other capabilities
- It has a **dedicated team of trained and experienced security analysts**, who can monitor and analyze incidents, respond processes, aggregate technologies, correlate and analyze data, and perform threat research and intelligence on an ongoing basis

Advantages

- This model helps an organization to meet specific **compliance requirements**
- It offers **cost-effective services** as compared to in-house SOC model
- It takes **less time** to build this model at an efficient level

Disadvantages

- It has the risk of external data mishandling
- It does not provide long-term gain to an organization

Types of SOC Models (Cont'd)

3. Hybrid SOC Model

- It is a combination of both **in-house** and **outsourced** SOC models
- In this model, the organization is accompanied with **MSSP** to offer the most secure approach

Advantages

- It shares **synergies** for technology, processes, expertise, facilities, and personnel to reduce the cost
- This model provides the best approach for **monitoring** and **analyzing intrusion incidents**, quick **detection** and **response** time, and **low backlogs**

Disadvantages

- It requires to sets up **extra** hardware, and managing data/information by third party
- It is **expensive** for long-term duration

Objective 05

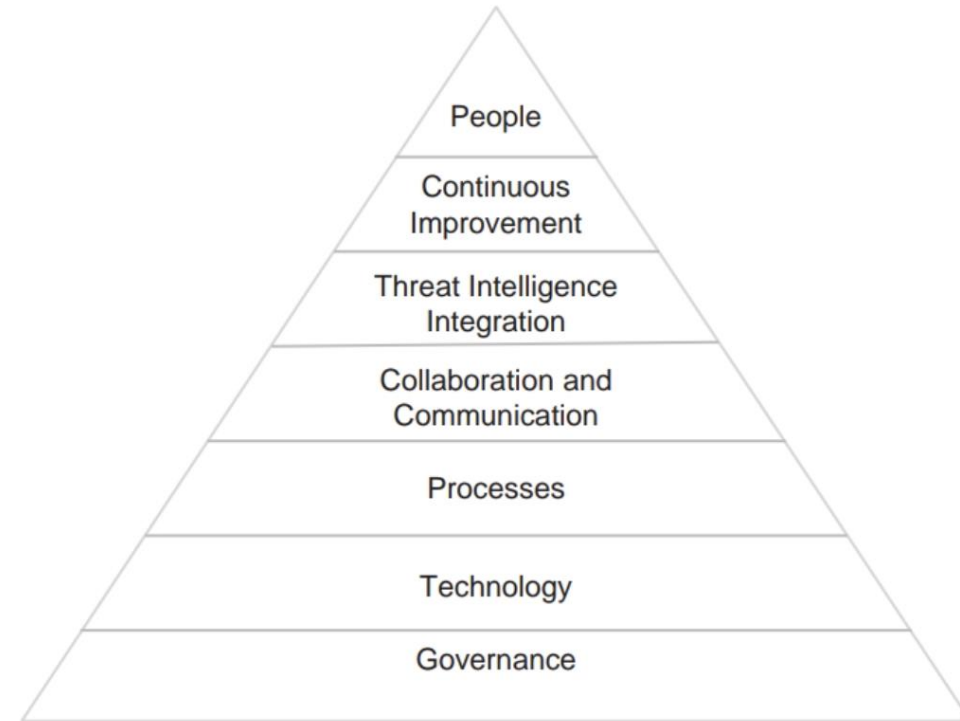
**Explain the Concept of SOC Maturity
Models and Evolution of SOC**

SOC Maturity Models

- Maturity models are **IT governance tools** that explain an organization's working process as per standardization, results, and measurement of effectiveness
- They are used to analyze where a **SOC succeeds** and where it requires improvements

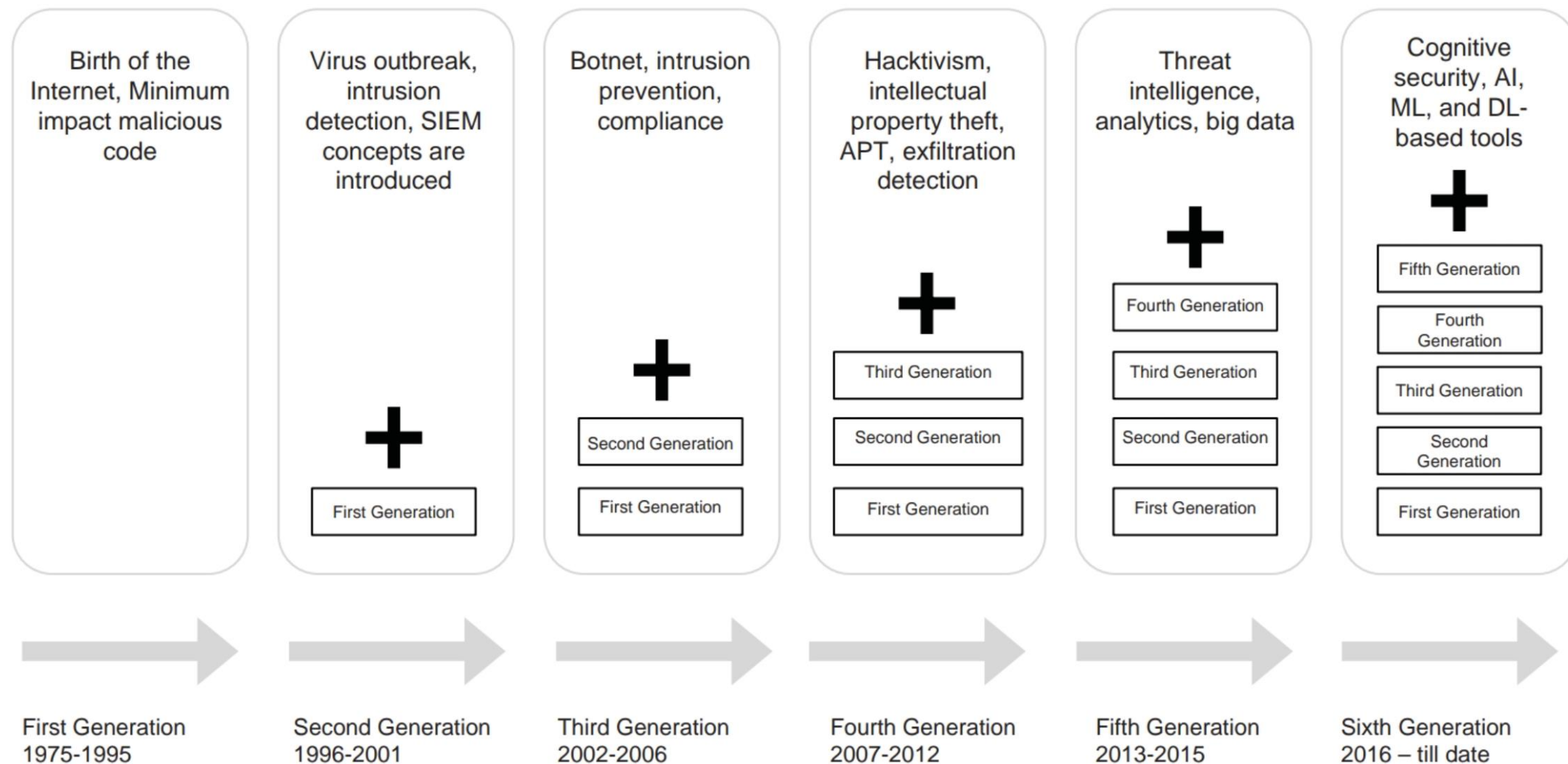
Examples of Maturity Models:

- 1 SOC-Capability Maturity Model
- 2 Control Objectives for Information Technology (COBIT)
- 3 National Institute of Standards and Technology (NIST) Cybersecurity Framework
- 4 Systems Security Engineering Capability Maturity Model (SSE-CMM)
- 5 Cybersecurity Capability Maturity Model (C2M2)



Components of SOC Maturity Models

SOC Generations



SOC Implementation

Planning	• Initial assessment is done based on existing capabilities with respect to people, processes, technology, business, and IT objectives • SOC strategy development is carried out by considering its strategic goal, scope, model, services, KPI, metrics, etc
Designing and Building the SOC	• Designing and building phases of SOC are almost linked to each other • Selection of the best technology to implement an efficient SOC is carried out in this phase
Operating the SOC	• SOC is moved from building phase to operating phase with the help of a proper transition plan
Reviewing & Reporting the SOC	• Reviews the SOC to identify the areas of improvement and to check whether it is operating accordingly

Objective 06

**Identify KPI's, Challenges and Implement
Best Practices for Effective SOC
Operations and Management**

Security Operations Center KPIs and Metrics

- Key Performance Indicators are quantifiable measurements used to evaluate the effectiveness of the Security Operations Center which are **Specific, Measurable, Actionable, Relevant** and **Timely**

SOC KPIs



Steps to Customize and Adapt Security Operations Metrics and Reporting



Challenges in Implementation of SOC

- 1 Increasing **volume** of security alerts
- 2 Management of **numerous** security tools
- 3 **Lack** of skilled analysts
- 4 **Legal** and **regulatory** compliance
- 5 Technology **selection** and **configuration**

- 6 Processes and procedures **formalization**, **orchestration**, and **automation**
- 7 **Data integrity** and **intelligence** management
- 8 **Handling** multi-staged advanced attacks
- 9 **Rapid change** in technology and security
- 10 Continuous training

Best Practices for Running SOC

- Always use real-time **security dashboards** for receiving notifications

- Use well-defined incident **response processes** and **procedures**

- Use **critical security alerts** to receive notifications 24x7

- Develop a **risk-aware environment** and **management system**

- **Use security intelligence** to manage security incidents

- Protect **mobile** and **social workspace**

- Receive **log** and **security** events from firewalls, routers, NetFlow, IPS, applications, and operating systems, for monitoring and analyzing

- Protect services, by **design**

- Automate security "hygiene"

- Ensure **resilience** by controlling access to the network

- Determine **complexity** of cloud and virtualization

- Maintain third-party **security compliance**

- Secure **data** and **protect** privacy

Improving the Effectiveness and Efficiency of SOC with AI

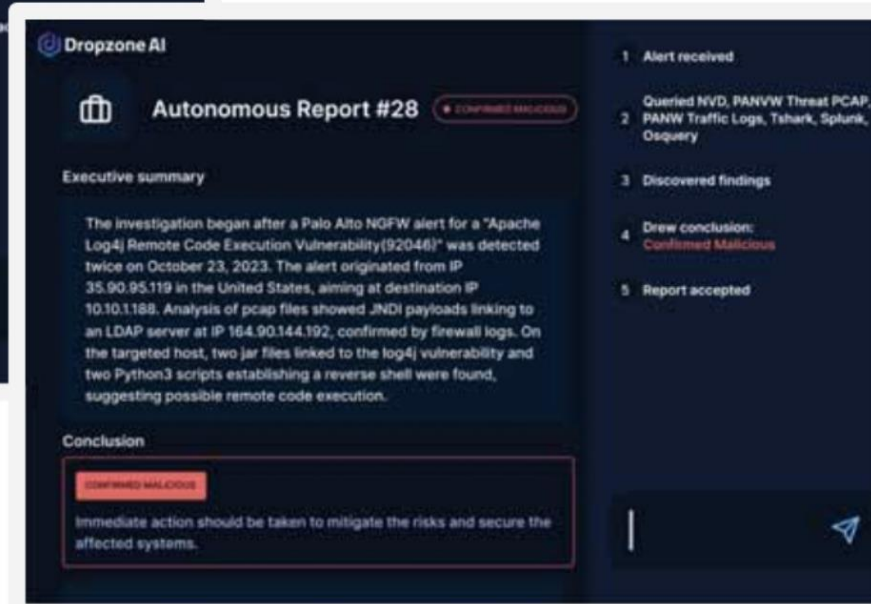
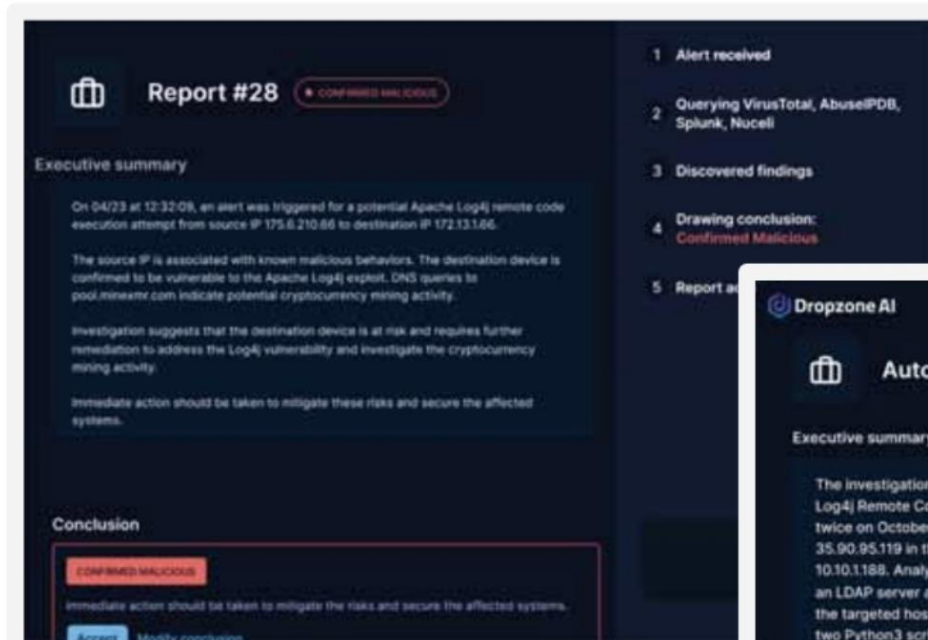
- AI-driven security operations center (SOC) leverages machine learning (ML) and artificial intelligence (AI) technologies to improve and streamline various aspects of security operations by integrating advanced analytics, automation, and intelligent decision-making processes

Key Services of AI-Driven SOC:

- Security Monitoring
- Security Incident Management
- Forensics & Analysis
- Threat Hunting
- Threat Intelligence

Key Features of AI-Driven SOC:

- Enhanced Alert Triage
- Automated and Intelligent Incident Response
- Enhanced Threat Detection
- Behavioural Analytics
- Predictive Analytics



Dropzone AI Driven SOC Dashboard

Source: www.dropzone.ai

Module **Summary**

- Security operation is the continuous operational practice for maintaining and managing a secure IT environment through the implementation and execution of certain services and processes
- A separate and dedicated unit is established by the organization to handle and manage their security operations, known as **Security Operation Center (SOC)**
- SOC provides a single point of view through which organization's security and assets are monitored, assessed, and defended
- A successful SOC operation is carried out in combination and coordination with **processes, technology, and people** working in the SOC
- A SOC analyst plays a pivotal role in SOC
- Outsourced SOC model, Managed Security Service Provider (MSSP) sets up the infrastructure and offers **threat intelligence** and other capabilities